Sector-wise wake losses of the Kelmarsh wind farm from a lattice Boltzmann large-eddy simulation on a laptop GPU

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Abstract

Wake losses of wind farms are usually estimated with engineering wake models, while large-eddy simulation (LES) is considered too expensive for routine sector-by-sector comparisons with operational data. This paper presents a compact lattice Boltzmann LES, written from published methods as a single module of GPU kernels in NVIDIA Warp, and applies it to the six-turbine Kelmarsh onshore wind farm (United Kingdom) on a laptop GPU with 6 GiB of memory. The turbines are actuator disks in a neutral atmospheric boundary layer fed by a concurrent periodic precursor. Seventy-two wind directions are simulated for each of two independent inflow realisations, and the resulting power table is evaluated on exactly the same 159 289 filtered ten-minute SCADA records (2016–2021) as three engineering models in PyWake. The farm wake loss is 9.81 % in the LES (two standard errors 0.11 %), 9.24 % measured (95 % interval 8.98–9.52 %) and 9.69 % with the Gaussian model of Niayifar and Porté-Agel. Over twelve 30-degree sectors the LES has the lowest root-mean-square error against the measurements, 2.2 percentage points against 2.6 for the Gaussian, 3.7 for the Jensen and 4.4 for the TurbOPark model, and the two inflow realisations agree within 0.52 points in every sector. All models overpredict the losses of the four easterly sectors, the LES by 2.9–4.5 points. A per-turbine decomposition traces this to a direction-dependent site effect on the free-stream reference turbine rather than to the wakes. One wind direction takes 81 s on the laptop GPU, a full sweep 1.7 GPU-hours. Code and result tables are openly available.

1 Introduction

Wake losses are one of the largest loss items in the energy yield of a wind farm, and they are almost always estimated with engineering wake models: the top-hat model of Jensen (1983) and Katić et al. (1986), Gaussian models (Bastankhah and Porté-Agel, 2014; Niayifar and Porté-Agel, 2016) or models designed for clusters of farms such as TurbOPark (Nygaard et al., 2020; Pedersen et al., 2022). They run in milliseconds, but their wake expansion and superposition rules carry empirical parameters, and different models applied to the same farm can differ by several percentage points of annual loss. Comparisons with operational data are therefore usually made per wind-direction sector, on measured power ratios of turbines in and out of wakes (Barthelmie et al., 2009).

Large-eddy simulation with actuator-disk or actuator-line rotors resolves the wakes and their interaction with atmospheric turbulence from physical inputs alone (Calaf et al., 2010; Stevens and Meneveau, 2017; Porté-Agel et al., 2020). The price is computational: a direction sweep of a real farm with turbulent inflow is usually a cluster job. The lattice Boltzmann method (LBM) is local and memory-bound, which suits graphics processors well, and LBM-based LES of wind-turbine wakes and of atmospheric boundary layers has been developed and assessed against established Navier–Stokes solvers (Asmuth et al., 2020, 2021).

This paper asks a practical question: can a consumer laptop GPU run an LES sweep of all wind directions of a real farm, evaluated in exactly the same way as engineering models against SCADA data, with an honest estimate of its own uncertainty? The test case is the Kelmarsh wind farm, whose SCADA data are openly published (Plumley and Takeuchi, 2025). The contributions are:

- a compact D3Q19 lattice Boltzmann LES with regularized collision, Smagorinsky closure, log-law wall model, filtered actuator disks and a concurrent precursor, verified against analytical solutions and momentum theory, in single and half precision;
- a direction sweep in which every turbine is referred to its own undisturbed inflow from a run without turbines, and whose power table goes through the same record selection, normalisation and sector aggregation as three PyWake models (Pedersen et al., 2019);
- an uncertainty estimate from two independent inflow realisations, checked against the scatter of ten-minute blocks within each realisation;
- a per-turbine decomposition of the sector losses, which locates the largest common discrepancy of all models in the free-stream reference rather than in the wakes;
- the cost on a 6 GiB laptop GPU, and openly available code and result tables.

2 Site and SCADA data

Kelmarsh consists of six Senvion MM92 turbines (rotor diameter $D = 92\,\mathrm{m}$, rated power 2.05 MW) in gently rolling farmland in Northamptonshire; turbines T3 and T6 have a hub height of 68.5 m, the other four 78.5 m. The dataset of Plumley and Takeuchi (2025) provides ten-minute SCADA records and the turbine positions under a CC-BY-4.0 licence. Ground elevation varies by about 20 m across the farm; no terrain model is used in this study.

The record selection and normalisation are those of an earlier engineering-model study of the same farm (Waleszczyński, 2026a), imported unchanged (pinned version) rather than re-implemented. Records from 2016 to 2021 are kept when all six turbines run in partial load (power above 20 kW, pitch below 5°) and report a nacelle direction; the wind direction is the circular mean of the six. For each 1-degree direction the free-stream turbines follow from the disturbed-sector rule of IEC 61400-12-1 (International Electrotechnical Commission, 2017). Each turbine carries a site factor c_i, its power relative to the other free-stream turbines over the records in which it is itself free, iterated to a fixed point with mean one; it removes the effect of hub height and position that holds in all directions. With $\hat{P}_i = P_i/c_i$, the reference power of a record is the mean of $\hat{P}_i$ over its free-stream turbines, and the reference wind speed follows from the inverse catalogue power curve. Records with a reference speed between 5 and 10 m/s are kept: 159 289 records on 1 913 days. The wake loss of a sector is

$$L = 1 - \frac{\sum_{\text{records}} \sum_{i=1}^{6} \hat{P}_i}{\sum_{\text{records}} 6\, P_{\text{ref}}}, \tag{1}$$

over 30-degree sectors centred on 0, 30, …, 330°. Its 95 % interval comes from a bootstrap of whole days (1000 resamples), because ten-minute records within a day are strongly correlated. The ambient turbulence intensity, the median ten-minute standard deviation of the nacelle anemometers over the mean speed, is 12.7 %.

The models are evaluated on the same records: each record is looked up in a power table $P_i(\mathrm{wd}, \mathrm{ws})$ at its own direction and reference speed, the model powers are normalised in the same way, and Eq. (1) is applied. The three PyWake models are the Jensen model with wake expansion $k = 0.075$, the Gaussian model of Niayifar and Porté-Agel (2016) and TurbOPark in the formulation of Pedersen et al. (2022), all with the measured ambient turbulence intensity, the catalogue power curve of the MM92 (Haas et al., 2024) and a generic thrust curve.

3 Numerical method

3.1 Lattice Boltzmann solver

The solver uses the D3Q19 velocity set (Krüger et al., 2017), with one fused kernel per time step (pull streaming, moments, forcing and collision) on two population arrays, written in NVIDIA Warp (Macklin, 2022); the same kernels compile for the CPU, which is how the automated tests run. The collision is regularized BGK (Latt and Chopard, 2006): the non-equilibrium part of the populations is projected on the second-order Hermite polynomials before relaxation, which keeps the scheme stable at the effectively infinite Reynolds number of the atmosphere,

$$f_q^{*} = f_q^{\mathrm{eq}} + \left(1 - \frac{1}{\tau}\right) f_q^{(1)} + \left(1 - \frac{1}{2\tau}\right) S_q, \tag{2}$$

with the forcing term S_q of Guo et al. (2002) for all body forces (driving pressure gradient, wall stress and rotor thrust). The first-order non-equilibrium moment $-\mathbf{F}/2$ has to be kept in $f^{(1)}$: a plain second-order projection drops it and then delivers only $(3-\omega)/2$ of the force, which the Poiseuille test below detected as 88 % of the expected flow rate at $\tau = 0.8$. The Smagorinsky eddy viscosity (Smagorinsky, 1963) enters the relaxation time and is computed locally from the non-equilibrium momentum flux Π (Hou et al., 1994),

$$\tau = \frac{1}{2}\left(\tau_0 + \sqrt{\tau_0^2 + 18\sqrt{2}\,(C_s\Delta)^2\|\Pi\|/\rho}\right), \qquad C_s = 0.16, \tag{3}$$

with the damping of Mason and Thomson (1992) near the ground. Populations are stored as deviations from the lattice weights, $f_q - w_q$, which keeps significant digits for the small deviations that carry the flow (Lehmann et al., 2022), in float32 or optionally float16.

3.2 Atmospheric boundary layer and inflow

The neutral surface layer is a pressure-driven open channel: a constant body force, a free-slip lid at 460 m and a rough ground represented by a log-law wall stress applied as a force in the first cell (local-velocity form, after Moeng, 1984),

$$\boldsymbol{\tau}_w = -\left[\frac{\kappa}{\ln(z_1/z_0)}\right]^2 |\mathbf{U}_1|\,\mathbf{U}_1, \qquad z_1 = \Delta/2, \quad \kappa = 0.4. \tag{4}$$

The roughness length $z_0 = 0.2$ m (farmland with hedgerows and woods) was chosen so that the hub-height turbulence intensity of the LES is close to the SCADA ambient value; it is the only parameter of the inflow that was adjusted to the site. Coriolis forces, veer and thermal stratification are not modelled.

The inflow turbulence comes from a periodic precursor, $12\,D$ long and as wide as the farm domain, whose wrap in the streamwise direction is shifted laterally so that streaks do not lock in place (Munters et al., 2016). It runs concurrently with the farm domain and hands over its outflow plane at every time step (Stevens et al., 2014); the farm domain has a zero-gradient outflow behind a $2\,D$ sponge layer. The precursor is spun up for 6000 s of physical time before any farm run.

The farm domain is the same for every wind direction: the turbines are rotated into a wind-aligned frame around the farm centre, with at least $5\,D$ between the inflow plane and the nearest possible rotor, $5\,D$ at the sides and $10\,D$ behind. Table 1 lists the main parameters. The LES is effectively Reynolds-number independent and the disks use a constant thrust coefficient, so a run at one hub-height speed represents all partial-load speeds; the lattice speed is mapped to 8 m/s.

3.3 Actuator disks

The rotors are thrust-only disks normal to the mean wind, smoothed with a Gaussian of width $\sigma = 0.75$ cells along the axis and an error-function edge in the rotor plane. The thrust uses the

Table 1: Simulation set-up of the main runs.

Grid spacing	$\Delta = D/8 = 11.5\,\mathrm{m}$ (D/12 and D/16 in the grid study)
Farm domain	$213 \times 173 \times 40$ cells ($2.45 \times 1.99 \times 0.46\,\mathrm{km}$)
Precursor	$96 \times 173 \times 40$ cells ($12\,D$ long), shifted periodic
Cells in total	2.14 million
Hub-height speed	0.05 in lattice units (Mach number 0.09), mapped to 8 m/s
Time step	0.069 s; about 35 000 steps per direction
Run	10 min spin-up, 30 min average in three 10-minute blocks
Directions	72 (every 5°), two inflow realisations
Disks	$C_T = 0.8$, $C_T' = 1.388$ (calibrated), Gaussian width $\sigma = 0.75\Delta$
Subgrid model	Smagorinsky, $C_s = 0.16$, Mason–Thomson wall damping
Ground	log-law wall model, $z_0 = 0.2\,\mathrm{m}$; free-slip lid at 460 m

disk-averaged velocity $\langle u_d \rangle$ of the previous step and the local thrust coefficient C_T' (Calaf et al., 2010), with the correction for filtered disks of Shapiro et al. (2019),

$$F = -\tfrac{1}{2}\rho\, C_T' M^2 \langle u_d \rangle^2 A, \qquad M = \left(1 + \frac{C_T'\,\Delta_f}{4\sqrt{3\pi}\,R}\right)^{-1}, \qquad \Delta_f = \sqrt{12}\,\sigma. \tag{5}$$

C_T' is calibrated per resolution so that an isolated disk in uniform inflow gives $C_T = 0.8$, the partial-load thrust coefficient of the generic thrust curve (no thrust curve of the MM92 is public). The calibration gives $C_T' = 1.388$ at $D/8$ and 1.427 at $D/12$, against 1.528 from momentum theory, and the ratio $r = \langle u_d \rangle / U_\infty = 0.814$ at $D/8$, which turns a disk velocity into a free-stream equivalent speed $\langle u_d \rangle / r$ for the power curve.

3.4 Runs and the reference without turbines

Each direction is a 10-minute spin-up followed by a 30-minute average, split into three SCADA-like ten-minute blocks. All directions of one inflow realisation start from the same spun-up precursor state and the solver is deterministic, so they see the same inflow turbulence (common random numbers), which suppresses most direction-to-direction sampling noise.

A 30-minute mean of the inflow is not uniform. Long-lived streaks leave the mean hub-height speed varying from 0.89 to 1.07 of the inflow hub speed across the domain (standard deviation 3.6 %; 0.90–1.03 and 2.3 % in the second realisation). Referring every turbine to the farm-wide inflow speed turned this into ±15 % of power and negative wake losses in some sectors. Because the inflow is identical in every run of a realisation, one extra run without turbines gives the undisturbed mean velocity at every possible rotor position. The equivalent speed ratio of turbine i in block b is therefore

$$s_{i,b} = \frac{\langle u_{d,i} \rangle_b}{r\,\langle u^0_{d,i} \rangle_b}\, h_i, \tag{6}$$

where u^0 is the same disk average in the run without turbines and h_i the shear factor of hub height i (68.5 or 78.5 m) relative to the reference height. With this reference the free-stream turbines run at 0.997 ± 0.013 of their own undisturbed inflow over 219 turbine-directions (0.998 ± 0.013 in the second realisation).

3.5 From speed ratios to wake losses

Because the flow scales with the wind speed, the LES gives the power table

$$P_i(\mathrm{wd}, \mathrm{ws}) = \frac{1}{N_b} \sum_b P_{\mathrm{curve}}(\mathrm{ws}\, s_{i,b}), \tag{7}$$

averaged over the $N_b = 6$ blocks of both realisations and interpolated linearly between the 5-degree directions. It enters the evaluation of Sect. 2 in place of a PyWake table. Sampling directions every

5 instead of 1 degree changes the sector losses of the Gaussian model by at most 0.18 points and its farm loss by less than 0.01 points.

The uncertainty of the LES loss is the standard error over blocks: the loss of Eq. (1) is evaluated with each block alone, and the standard deviation of the six results is divided by $\sqrt{6}$. This treats the three blocks of one realisation as independent, which the second realisation tests (Sect. 5.3).

4 Verification

Table 2 and Fig. 1 summarise the checks. The viscosity recovered from the decay of a Taylor–Green vortex converges at second order, the body-force-driven Poiseuille flow is reproduced to about 1 %, and an actuator disk in uniform inflow follows momentum theory closely at both resolutions; at a local thrust coefficient of 1.5 the induction is 0.255 at $D/8$ and 0.262 at $D/16$ against 0.273 from theory. The precursor reproduces the log law in the first cell (mean speed 8.45 against 8.40 u_*), while the hub-height speed lies 4 % below it, the log-layer mismatch typical of wall-modelled LES with a Smagorinsky closure (Brasseur and Wei, 2010). The resolved shear stress reaches $0.93\,u_*^2$ at 40 m, and the hub-height turbulence intensity is 13.4 % (12.9 % in the second realisation) against 12.7 % from the SCADA.

Table 2: Verification and sensitivity checks.

Check	Result
Taylor–Green vortex, viscosity from energy decay, 16/32/64/128 cells per wavelength	error 10.0 / 2.5 / 0.70 / 0.29 %, second order
Poiseuille channel (body force, bounce-back walls), 8/16/32 cells, $\tau = 0.8$	maximum profile error 1.1 / 1.1 / 0.23 %
Actuator disk in uniform inflow, $C_T' = 1.5$, $D/8$ and $D/16$	induction 0.255 / 0.262 (theory 0.273); C_T 0.832 / 0.817 (theory 0.793)
Inflow profile, $D/8$	first cell 8.45 u_* (log law 8.40); hub 4 % below the log law; resolved shear stress 0.93 u_*^2 at 40 m
Inflow turbulence at hub height	TI 13.4 % and 12.9 % (SCADA 12.7 %); $\sigma_u, \sigma_v, \sigma_w$ = 1.9, 1.4, 1.1 u_*
Direction step of 5° instead of 1° (Gaussian model)	sector losses change by at most 0.18 pp, farm loss by less than 0.01 pp
Grid, 240° and 300°, $D/8$, $D/12$, $D/16$	loss at 8 m/s: 19.5 / 18.5 / 20.5 % and 30.5 / 28.8 / 29.6 %; hub TI 13.4 / 12.4 / 12.5 %
float16 populations	Taylor–Green error 0.76 / 0.60 % at 64 / 128 cells; farm loss at 240° 19.9 against 19.5 %, every speed ratio within 0.017 of float32

Grid study. The two directions with the deepest wakes, 240° and 300°, were repeated at 12 and 16 cells per rotor diameter, each resolution with its own disk calibration and precursor. The farm loss of one direction at 8 m/s (farm power over six times the mean power of the free-stream turbines) is 19.5, 18.5 and 20.5 % at 240° and 30.5, 28.8 and 29.6 % at 300° for $D/8$, $D/12$ and $D/16$. The changes are within one to two standard errors of the ten-minute blocks (0.2–1.2 points) and show no trend. Because every resolution has its own turbulence realisation, the study cannot resolve differences below about one point.

Half precision. With float16 populations the Taylor–Green errors stay below 1 % and the farm loss at 240° changes from 19.5 to 19.9 %, each turbine's speed ratio within 0.017 of its float32 value. Half precision halves the memory and raises the kernel throughput by about 6 % (Sect. 5.6); all results below use float32.

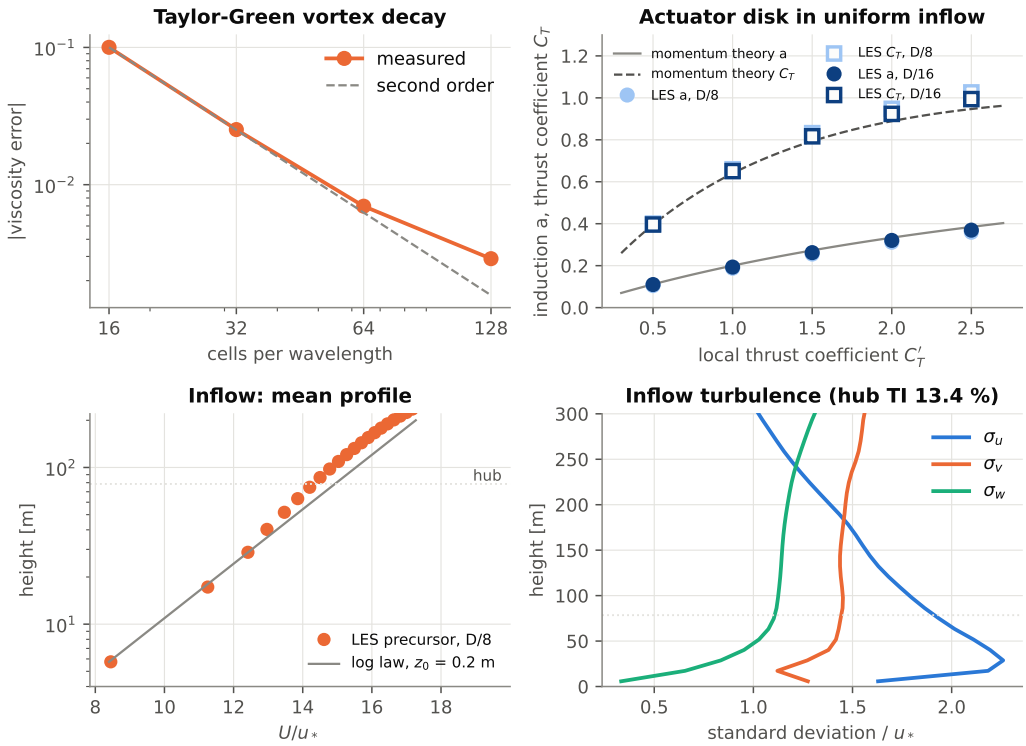


Figure 1: Verification. Top left: viscosity error of the Taylor–Green vortex. Top right: induction and thrust coefficient of an isolated actuator disk against momentum theory. Bottom: mean velocity and turbulence profiles of the precursor at $D/8$ (dotted line: hub height).

5 Results

5.1 Flow field

Figure 2 shows the hub-height flow for the two directions with the deepest wakes. At 300° the turbines stand in two rows, T3–T5–T6 and T2–T1–T4, whose wakes merge downstream; at 240° three pairs stand one behind the other (T2–T3, T1–T5, T4–T6). The instantaneous fields show the wakes meandering in the streaky inflow.

5.2 Wake loss per sector

Table 3 and Fig. 3 compare the wake losses. The LES farm loss, 9.81 %, lies 0.6 points above the measured 9.24 % and close to the Gaussian model (9.69 %), while Jensen and TurbOPark overpredict by 2.2 and 2.9 points. Over the twelve sectors the LES has the lowest root-mean-square error and the smallest mean bias. The deep-wake sectors are reproduced: along the row T3–T5–T6 at 300° the LES gives 22.1 % against 22.2 % measured, where the Gaussian model is 1.9 points low and Jensen and TurbOPark 1.3 and 2.8 points high. At 240°, the most frequent sector, the LES is 1.1 points low. Figure 4 resolves the farm efficiency in 5-degree bins.

5.3 Inflow realisations and the uncertainty of the LES

The second realisation repeats the whole chain with a precursor started from a different random seed, its own run without turbines and the same 72 directions. Its farm loss is 9.83 % against 9.79 % for the first, its sector-wise RMSE and bias against the SCADA are the same, and no sector differs by more than 0.52 points (root mean square over the sectors 0.24 points). If the ten-minute blocks of a realisation were less independent than assumed, the two realisations would differ by more than their block scatter predicts; the block standard errors predict a root-mean-square difference of 0.44 points, larger than observed. The block-based interval, 0.2–0.7 points per sector and 0.11 points on

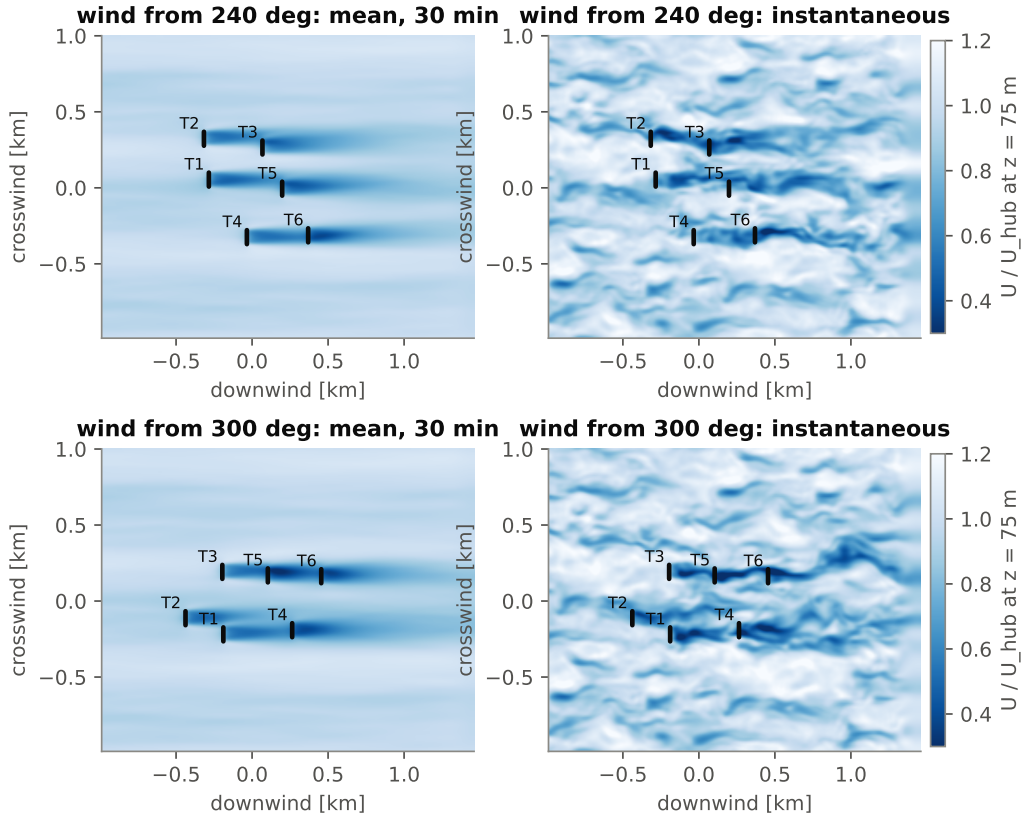


Figure 2: Streamwise velocity at $z = 75\,\mathrm{m}$, the grid level nearest to the 78.5 m hub height, for wind from 240° (top) and 300° (bottom): 30-minute mean (left) and an instantaneous field (right), normalised by the inflow hub speed.

Table 3: Wake loss per 30-degree sector [%] on the same SCADA records. SCADA with its 95 % day-bootstrap interval; LES pooled over two inflow realisations with two standard errors of its six ten-minute blocks, and each realisation on its own (r0 / r1).

Sector	Records	SCADA [95 %]	LES ± 2 SE	r0 / r1	Gaussian	Jensen	TurbOPark
0°	7 120	5.4 [5.0–5.7]	4.8 ± 0.3	4.8 / 4.7	5.2	6.0	6.4
30°	10 791	3.3 [2.7–3.8]	3.7 ± 0.3	3.6 / 3.9	4.9	6.0	5.8
60°	7 733	7.8 [6.9–8.6]	10.7 ± 0.3	10.8 / 10.6	11.0	13.3	14.5
90°	6 041	2.1 [1.4–2.9]	6.6 ± 0.3	6.6 / 6.7	7.4	8.6	9.8
120°	4 622	19.7 [18.8–20.7]	22.8 ± 0.5	22.7 / 22.9	20.9	24.4	25.4
150°	6 606	5.9 [5.5–6.3]	9.6 ± 0.3	9.7 / 9.4	10.3	11.7	12.3
180°	14 731	3.7 [3.4–4.1]	5.2 ± 0.4	4.9 / 5.4	5.7	6.6	7.1
210°	28 137	3.9 [3.7–4.1]	5.6 ± 0.2	5.7 / 5.5	5.4	6.5	6.5
240°	29 728	14.4 [14.0–14.8]	13.3 ± 0.7	13.3 / 13.3	12.4	14.9	16.1
270°	19 019	8.2 [7.9–8.5]	7.1 ± 0.4	7.1 / 7.1	7.4	8.9	9.5
300°	16 219	22.2 [21.7–22.7]	22.1 ± 0.6	22.1 / 22.1	20.3	23.5	25.0
330°	8 542	9.0 [8.7–9.4]	9.7 ± 0.4	9.6 / 9.8	10.9	12.2	12.7
all	159 289	9.24 [8.98–9.52]	9.81 ± 0.11	9.79 / 9.83	9.69	11.44	12.13
RMSE			2.2	2.2 / 2.2	2.6	3.7	4.4
bias			+1.3	+1.3 / +1.3	+1.4	+3.1	+3.8

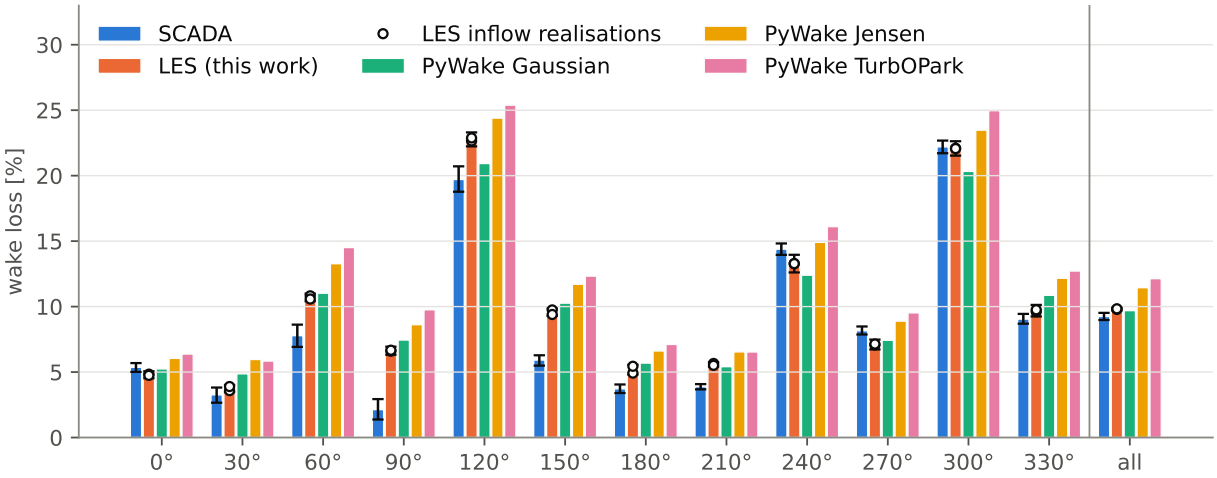


Figure 3: Wake loss per sector. Whiskers: SCADA 95 % day-bootstrap interval; LES two standard errors of its six ten-minute blocks from two inflow realisations. Circles: the two realisations on their own.

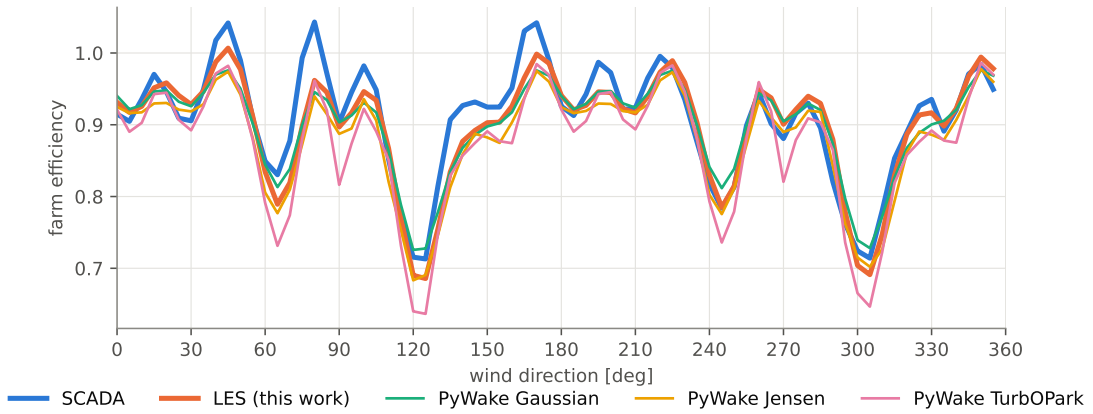


Figure 4: Farm efficiency (one minus the wake loss) in 5-degree bins with at least 100 records. The measured curve rises above one in a few directions, a site effect that the site factors only partly remove; at 60–80° and 130–160° every model stays below the measurement.

the farm loss, is therefore not optimistic. Two realisations are enough for this check but not to estimate the between-realisation spread on its own.

5.4 Turbine by turbine: the easterly sectors

Every model, the LES included, overpredicts the losses of the four easterly sectors (60–150°), the LES by 2.9–4.5 points. Because the farm loss of a sector is the mean of the six turbine losses against the same reference, Eq. (1) splits exactly into turbines. Averaged over the four sectors, most of the LES excess comes from T2 (11.3 points of turbine loss) and T1 (5.4 points); T3, T4 and T5 are within 2.4 points and T6 is 1.9 points below the measurement. The Gaussian model shows the same pattern (T2 9.6, T1 6.7, T6 −3.5 points).

Figure 5 shows why. The wake minima agree: at 140° T2 produces 0.33 of the reference power in the SCADA and 0.34 in the LES. What differs is the level outside the wakes. In the 5-degree bins where neither the LES nor the Gaussian model sees a wake on a turbine (Table 4), the measured turbines T2 and T3 produce 6 and 5 % more than the free-stream reference in easterly winds and T6 6 % less, while in westerly winds T1–T5 are within 1 % of it. T6 is one of the reference turbines in every easterly direction, so the reference itself is low there and the measured losses come out small. A single site factor per turbine cannot remove such an effect: T6 is 6 % low in easterly and 7 % high in westerly winds, and its factor averages the two. A direction-dependent effect of this size is absent from flat-terrain models; terrain, roughness changes or obstacles upwind of T6 are plausible causes that this study cannot separate.

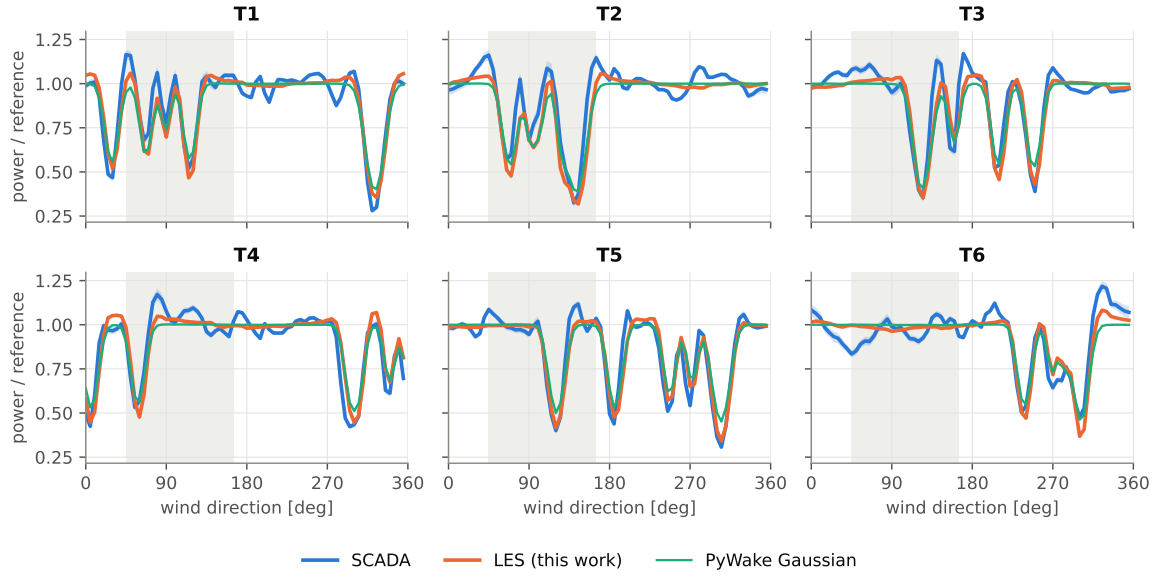


Figure 5: Power of each turbine relative to the free-stream reference in 5-degree bins (SCADA with its 95 % interval over days). Shaded: the easterly sectors 60–150°.

Table 4: Power of unwaked turbines relative to the free-stream reference: record-weighted mean over the 5-degree bins with at least 300 records in which neither the LES nor the Gaussian model sees a wake on the turbine (ratio above 0.99).

	T1	T2	T3	T4	T5	T6
SCADA, easterly winds (0–175°)	1.01	1.06	1.05	1.03	1.02	0.94
SCADA, westerly winds (180–355°)	1.01	1.00	1.00	1.00	1.00	1.07
LES, easterly winds	1.02	1.03	1.02	1.02	1.00	1.00
LES, westerly winds	1.01	1.00	1.01	1.00	1.02	1.02

5.5 Single wake

Behind a single turbine in the same inflow (Fig. 6) the LES centreline deficit, relative to the same position in the run without turbines, is deeper than in the Gaussian and Jensen models up to about $8\,D$ (0.32 against 0.26 and 0.22 at $4\,D$) and recovers to 0.11 at $10\,D$. The TurbOPark centreline deficit in PyWake stays much deeper (0.46 at $4\,D$), consistent with its overprediction of the farm loss.

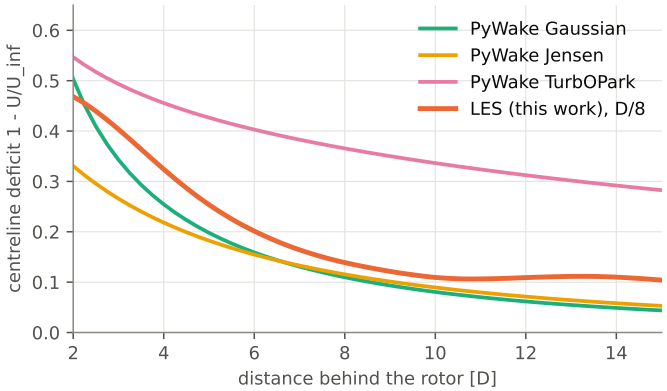


Figure 6: Hub-height centreline velocity deficit behind a single turbine: LES at $D/8$ and the PyWake models at the LES inflow turbulence intensity.

5.6 Computational cost

The laptop GPU is an NVIDIA Quadro RTX 3000 (Turing, 6 GiB). The bare float32 kernel runs at 1240–1320 million lattice updates per second (MLUPS) from 0.3 to 19 million cells (2.7 GiB of populations); with float16 storage it reaches 1340 MLUPS and fits 37 million cells in 2.6 GiB (Fig. 7). A farm run, farm domain and precursor together with 2.14 million cells, runs at about 920 MLUPS including disk forces and averaging: 81 s per wind direction, 1.7 GPU-hours for 72 directions and below 1.5 GiB of GPU memory. Each further realisation costs the same again. At $D/12$ and $D/16$ one direction takes 413 and 1290 s with 2.1 and 3.6 GiB, so a full sweep at $D/16$ would take about 26 GPU-hours.

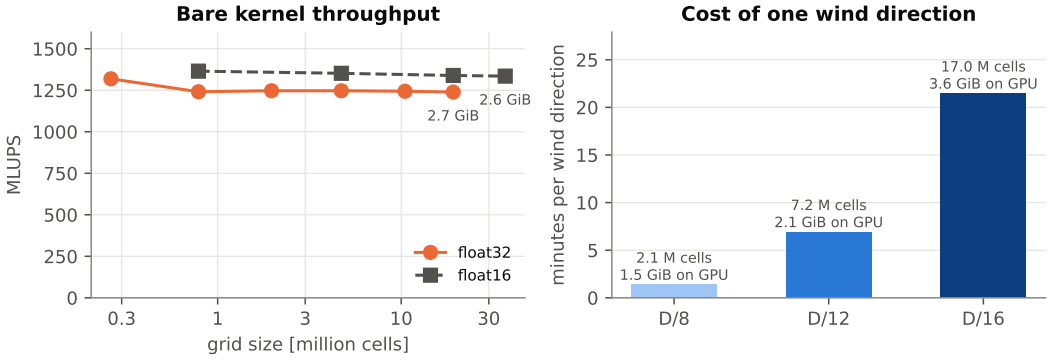


Figure 7: Left: throughput of the bare kernel against grid size (labels: memory of the populations). Right: wall time of one wind direction (10-minute spin-up and 30-minute average) at three resolutions.

6 Limitations

- *Neutral, flat, no Coriolis.* No veer, stability or terrain. The easterly sectors show a direction-dependent site effect on the free-stream turbines that none of the models contains.

- *Coarse rotors.* At $D/8$ the rotor spans eight cells and the hub lies seven cells above the ground. The disks are calibrated to the right thrust, and up to $D/16$ the loss changes by about one point without a trend, but the grid study covers two directions, cannot separate resolution from turbulence realisation below that level and does not resolve the near wake.
- *One wind speed.* The LES runs with $C_T = 0.8$ and is scaled to the 5–10 m/s records; the generic thrust curve falls to 0.71 at 10 m/s, so the fastest records are slightly over-thrusted.
- *Thrust-only disks, perfectly yawed.* No rotation, nacelle or tower, no yaw misalignment.
- *Two inflow realisations.* Enough to test the block-based uncertainty, not to estimate the between-realisation spread with precision.
- *Different averaging.* The LES turbulence intensity is a 30-minute standard deviation of the resolved field, the SCADA value the median ten-minute standard deviation of nacelle anemometers.

7 Conclusions

A lattice Boltzmann LES written from published methods runs a complete 72-direction sweep of a six-turbine onshore farm in 1.7 hours on a 6 GiB laptop GPU. Evaluated on exactly the same SCADA records and with the same normalisation as three engineering wake models, it gives a farm wake loss of 9.81 % against 9.24 % measured and the lowest sector-wise error of the four, without a tunable wake parameter: its inputs are a roughness length chosen for the inflow turbulence, a thrust coefficient and the turbine positions. Two ingredients made the comparison meaningful at this cost: referring every turbine to its own undisturbed inflow from a run without turbines, which removes the bias of long-lived streaks, and a second inflow realisation, which showed that the block-based uncertainty is not optimistic. The largest common error of all models, in the easterly sectors, is not a wake error: the per-turbine decomposition places it in a direction-dependent site effect on the free-stream reference turbine, which a single site factor per turbine cannot remove. For SCADA-based wake-loss assessments this suggests checking the unwaked levels of the reference turbines direction by direction before attributing a model discrepancy to the wake model.

Code and data availability

The solver, the analysis and all result tables are available under the MIT licence at `https://github.com/kwwaleszczynski-del/wind-les-gpu` (Waleszczyński, 2026b); the numbers in this paper correspond to commit 3fe934a. The SCADA record selection and the PyWake models are taken from Waleszczyński (2026a) (commit 2bb9f11). The Kelmarsh data are available from Zenodo (Plumley and Takeuchi, 2025) under CC-BY-4.0.

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Competing interests

The author declares no competing interests.

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